Important Questions with Answers

Chapter 2: Shaping of the Earth's Surface
Class 9 Social Science (NCERT) - Understanding Society: India and Beyond, Part 1

A. MCQs B. Fill in the Blanks C. True/False D. Match the Following E. Assertion-Reason F. Short Answer G. Long Answer

A. Multiple Choice Questions (1 mark each)

Q1. A natural feature on the Earth's surface formed by weathering, erosion, deposition or movement of the crust is called a:

(a) Plate
(b) Landform
(c) Mantle
(d) Boundary

Answer: (b) Landform

Q2. According to the chapter, the theory of plate tectonics was given by:

(a) Varahamihira
(b) Kautilya
(c) W.J. Morgan
(d) Anna Mani

Answer: (c) W.J. Morgan

Q3. The crust together with the upper part of the mantle forms the:

(a) Asthenosphere
(b) Lithosphere
(c) Outer core
(d) Inner core

Answer: (b) Lithosphere

Q4. The semi-molten layer beneath the lithosphere that allows plates to move is the:

(a) Asthenosphere
(b) Crust
(c) Inner core
(d) Atmosphere

Answer: (a) Asthenosphere

Q5. The movement of tectonic plates is caused by:

(a) Wind currents
(b) Convection currents in the mantle
(c) Ocean tides
(d) Rainfall

Answer: (b) Convection currents in the mantle

Q6. The Himalaya were formed by the collision of two:

(a) Oceanic plates
(b) Continental plates
(c) Transform plates
(d) Divergent plates

Answer: (b) Continental plates (convergent boundary, forming fold mountains)

Q7. The Mid-Atlantic Ridge is an example of a:

(a) Convergent boundary
(b) Divergent boundary
(c) Transform boundary
(d) Mixed boundary

Answer: (b) Divergent boundary

Q8. The San Andreas Fault in the United States lies along a:

(a) Convergent boundary
(b) Divergent boundary
(c) Transform boundary
(d) Glacier

Answer: (c) Transform boundary

Q9. The area around the Pacific Ocean where most earthquakes and volcanoes occur is called the:

(a) Ring of Fire
(b) Rift Valley
(c) Karst Region
(d) Great Divide

Answer: (a) Ring of Fire

Q10. India's only mud volcano is located at:

(a) Barren Island
(b) Baratang Island
(c) Lakshadweep
(d) Sundarbans

Answer: (b) Baratang Island (Andaman and Nicobar Islands)

Q11. Breaking down of rocks into smaller pieces WITHOUT movement of the material is called:

(a) Erosion
(b) Deposition
(c) Weathering
(d) Transportation

Answer: (c) Weathering

Q12. When plant roots grow into cracks and split rocks apart, it is an example of:

(a) Physical weathering
(b) Chemical weathering
(c) Biological weathering
(d) Glacial erosion

Answer: (c) Biological weathering

Q13. Which is a water conservation technique of digging trenches along the contour lines of a hillside?

(a) Terracing
(b) Contouring
(c) Bunding
(d) Check dam

Answer: (b) Contouring (continuous contour trench)

Q14. U-shaped valleys are carved by:

(a) Rivers
(b) Wind
(c) Glaciers
(d) Waves

Answer: (c) Glaciers

Q15. A winding bend in the middle or lower course of a river is called a:

(a) Meander
(b) Delta
(c) Waterfall
(d) Cirque

Answer: (a) Meander

Q16. Sea arches and sea stacks are formed by the action of:

(a) Glaciers
(b) Wind
(c) Waves
(d) Groundwater

Answer: (c) Waves (coastal erosion)

Q17. Crescent-shaped dunes formed in areas of limited sand and a single wind direction are:

(a) Star dunes
(b) Barchan dunes
(c) Parabolic dunes
(d) Longitudinal dunes

Answer: (b) Barchan dunes

Q18. Icicle-shaped formations hanging from the ceiling of a cave are called:

(a) Stalagmites
(b) Stalactites
(c) Sinkholes
(d) Dolines

Answer: (b) Stalactites

Q19. GLOF stands for:

(a) Global Land Outflow
(b) Glacial Lake Outburst Flood
(c) Great Lava Outflow
(d) Glacial Land Overflow

Answer: (b) Glacial Lake Outburst Flood

Q20. Which of the following is a human factor that causes landslides?

(a) Heavy rainfall
(b) Earthquakes
(c) Deforestation and unplanned construction on hillsides
(d) Steep slopes

Answer: (c) Deforestation and unplanned construction on hillsides

B. Fill in the Blanks (1 mark each)

Q21. The Earth is made up of three main layers: crust, mantle and _____.

Answer: core

Q22. Tectonic plates move very slowly, usually a few _____ per year.

Answer: centimetres

Q23. Plates that carry both continents and oceans are called _____ plates.

Answer: mixed

Q24. At a _____ boundary, plates move away from each other and magma rises to form new crust.

Answer: divergent

Q25. In early times, earthquakes were known as '_____', meaning shaking of the Earth.

Answer: bhukampa

Q26. Varahamihira dedicated a section of the *Brihatsamhita* to _____.

Answer: earthquakes

Q27. The process by which soil and rocks are carried away from one place to another by water, wind, ice or waves is called _____.

Answer: erosion

Q28. Earthen embankments built along contour lines to slow surface run-off are called _____.

Answer: bunding

Q29. The Zabo system is an integrated farming approach practised in _____.

Answer: Nagaland

Q30. In the upper course, rivers form V-shaped valleys, waterfalls and _____.

Answer: rapids

Q31. Oxbow lakes and floodplains are formed in the _____ course of a river.

Answer: middle

Q32. Rock, soil and debris carried and left behind by glaciers is called _____.

Answer: till

Q33. Streamlined rock ridges carved by wind are called _____.

Answer: yardangs

Q34. Depressions formed when the ground collapses into an underground cavity are called _____ or dolines.

Answer: sinkholes

Q35. Dust storms are common in desert and _____ regions where the soil is loose and dry.

Answer: semi-arid

C. True or False (1 mark each)

Q36. The Earth's surface is constant and does not change.

Answer: False. It is constantly being transformed by forces acting from within and on the surface.

Q37. Continental crust is thicker (about 30 km) than oceanic crust (about 5 km).

Answer: True.

Q38. The inner core is a solid, hot, spinning metal ball and is the densest part of the Earth.

Answer: True.

Q39. The outer core is mainly made of iron and nickel and is fluid.

Answer: True.

Q40. At a transform boundary, plates collide and form fold mountains.

Answer: False. At a transform boundary, plates slide past each other; collision happens at a convergent boundary.

Q41. When an oceanic plate collides with a continental plate, the oceanic plate sinks beneath it, causing volcanic activity and earthquakes.

Answer: True.

Q42. Weathering involves movement of the broken material.

Answer: False. Weathering only breaks rocks down; movement of material is erosion.

Q43. Erosion can destroy the fertile topsoil that farmers depend on.

Answer: True.

Q44. Waterfalls form where hard rock resists erosion and softer rock below is worn away.

Answer: True.

Q45. Deltas are formed in the upper course of a river.

Answer: False. Deltas are formed at the river's mouth, where it enters a sea, ocean or lake.

Q46. Moraines are formed by deposition of material carried by glaciers.

Answer: True.

Q47. Wind erosion is common in dry and sandy areas.

Answer: True.

Q48. Stalagmites hang from the cave ceiling, while stalactites rise from the cave floor.

Answer: False. It is the opposite: stalactites hang from the ceiling and stalagmites rise from the floor.

Q49. Avalanches can be triggered by a sudden rise in temperature or heavy snowfall.

Answer: True.

Q50. Landforms have influenced human history, e.g. the Himalayas protected India while passes like the Khyber allowed cultural exchange.

Answer: True.

D. Match the Following (1 mark per pair)

Q51. Match the plate boundary with its feature or example.

	Column A		Column B
1	Convergent boundary	A	Mid-Atlantic Ridge
2	Divergent boundary	B	Most earthquakes and volcanoes around the Pacific
3	Transform boundary	C	Plates slide past each other, e.g. San Andreas Fault
4	Ocean-continent collision	D	Himalayan fold mountains (collision of plates)
5	Ring of Fire	E	Oceanic plate sinks beneath continental plate, causing volcanoes

Answer: 1-D; 2-A; 3-C; 4-E; 5-B

Q52. Match the agent of gradation with the landform it creates.

	Column A		Column B
1	Running water	A	Sea cliffs, beaches and bays
2	Glaciers	B	Deltas and meanders
3	Wind	C	Caves, sinkholes and stalactites
4	Waves	D	U-shaped valleys and cirques
5	Groundwater	E	Dunes and yardangs

Answer: 1-B; 2-D; 3-E; 4-A; 5-C

Q53. Match the landform with its description.

	Column A		Column B
1	Cirque	A	Rock polished and shaped by sandblasting
2	Arete	B	Deep, narrow inlet formed when sea floods a glacial valley
3	Fjord	C	Bowl-shaped depression at the head of a glacier
4	Ventifact	D	Lake formed from a cut-off river bend
5	Oxbow lake	E	Sharp ridge between valleys

Answer: 1-C; 2-E; 3-B; 4-A; 5-D

E. Assertion-Reason Questions (1 mark each)

Options: *(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.*

Q54. Assertion (A): Tectonic plates move over the Earth's surface.
Reason (R): Convection currents in the mantle push and pull the plates.

Answer: (a)

Q55. Assertion (A): Most earthquakes and volcanoes occur along plate boundaries.
Reason (R): Plates interact, collide, separate or slide past each other at their boundaries.

Answer: (a)

Q56. Assertion (A): Weathering and erosion are the same process.
Reason (R): Both involve movement of broken rock material.

Answer: (d) A is false; R is false for weathering. Weathering only breaks down rock; erosion involves movement.

Q57. Assertion (A): Deltas are ideal for agriculture.
Reason (R): Deltas are formed of rich alluvial soil deposited by rivers.

Answer: (a)

Q58. Assertion (A): Glaciers form U-shaped valleys.
Reason (R): Glaciers widen and deepen existing river valleys as they move.

Answer: (a)

Q59. Assertion (A): Deforestation increases the risk of landslides.
Reason (R): Tree roots hold the soil together and removal of trees destabilises slopes.

Answer: (a)

Q60. Assertion (A): Sand dunes can protect settlements.
Reason (R): Dunes act as natural barriers against desertification, wind erosion and strong sea winds.

Answer: (a)

Q61. Assertion (A): A GLOF can occur when an earthquake or landslide strikes a glacial lake.
Reason (R): Glacial lakes are dammed only by hard, solid rock that can never collapse.

Answer: (c) A is true; R is false. Their natural dams are made of ice or loose moraines, which can weaken and collapse.

F. Short Answer Questions (2 to 3 marks each)

Q62. What is a landform? Give examples.

Answer: A landform is a natural feature on the Earth's surface formed by processes such as weathering, erosion, deposition and movement of the crust. Examples: mountains, valleys, plateaus, plains, deserts and coastal features.

Q63. What is plate tectonics?

Answer: Plate tectonics is a theory that explains the movement of the Earth's crust. The outermost layer is broken into large and small pieces called tectonic plates, which move slowly over the semi-molten layer beneath them and cause mountains, earthquakes and volcanoes.

Q64. Name the three layers of the Earth and describe the crust.

Answer: The three layers are the crust, mantle and core. The crust is the outermost layer on which we live; its thickness varies from 30-40 km under continents to 5-7 km under the ocean.

Q65. Differentiate between lithosphere and asthenosphere.

Answer: The lithosphere is the rigid outer layer including the crust and upper mantle (about 100 km). The asthenosphere is the hot, mobile, partially molten layer beneath it that allows plates to move.

Q66. Name the three types of tectonic plates and the major plates of the world.

Answer: Continental plates (carry continents), oceanic plates (carry ocean floors) and mixed plates (carry both). Major plates: Pacific, Eurasian, African, North American, South American, Indo-Australian and Antarctic.

Q67. How do convection currents move the plates?

Answer: Heat from the Earth's core makes molten material in the mantle rise while cooler material sinks. This continuous movement creates convection currents that push and pull the plates in different directions.

Q68. What are the three types of plate boundaries? Give one example of each.

Answer: Convergent (plates move towards each other) - Himalaya; divergent (plates move apart) - Mid-Atlantic Ridge; transform (plates slide past each other) - San Andreas Fault.

Q69. What is the Ring of Fire?

Answer: It is the area around the Pacific Ocean where most of the world's earthquakes and volcanoes occur along plate boundaries.

Q70. What did Varahamihira write about earthquakes?

Answer: In the Brihatsamhita he noted that changes in wind, rain, clouds, animal behaviour and planetary alignments could signal earthquakes. He attributed them to four elemental forces: Vayu (wind), Agni (fire), Indra (heaven/thunder) and Varuna (water). It was an early blend of observation and cosmological reasoning.

Q71. What is the mud volcano at Baratang Island?

Answer: It is India's only mud volcano, in the Andaman and Nicobar Islands. Unlike fiery volcanoes, mud bubbles out because of natural underground gases and pressure.

Q72. Name the three types of weathering with one example each.

Answer: Physical weathering - rocks break due to temperature change, frost or wind; chemical weathering - minerals change through reaction with water, air or acids (e.g. rain acting on rock); biological weathering - plant roots split rocks.

Q73. Name the types of erosion.

Answer: Water erosion (rivers, rain, waves), wind erosion, glacial erosion and coastal erosion.

Q74. How does erosion affect human occupations?

Answer: It removes fertile topsoil and lowers yields for farmers, washes away land, houses and roads near rivers and coasts, destabilises land in construction and mining, and damages tourism and fishing.

Q75. Explain contouring, bunding and terracing.

Answer: Contouring: trenches along the contour lines of a hillside to slow, hold and infiltrate rainwater. Bunding: earthen embankments along contour lines to slow run-off and increase soil moisture. Terracing: a series of level or gently sloping steps on a hillside to prevent soil erosion.

Q76. What are agents of gradation?

Answer: Natural forces that wear down, transport and deposit material on the Earth's surface and level it over time: running water, glaciers, wind, waves and groundwater.

Q77. What is a delta? Why is it important for humans?

Answer: A delta is a fan-shaped or triangular landform formed where a river meets a sea, ocean or lake and deposits sediment. It has rich alluvial soil good for rice and jute, supports fishing, trade and dense settlements, but is prone to flooding.

Q78. What is a meander? Why is it useful?

Answer: A meander is a winding bend in the middle or lower course of a river formed by erosion on outer banks and deposition on inner banks. The fertile soil supports agriculture, and meanders help navigation, irrigation and tourism.

Q79. What are moraines? Name their types.

Answer: Moraines are landforms formed by deposition of rocks, soil and debris (till) left behind by glaciers. Types: lateral, medial and terminal moraines.

Q80. What are dunes? Name any three types.

Answer: Dunes are hills or ridges of sand formed by wind in deserts or along sandy coasts. Types: barchan, longitudinal, star and parabolic dunes.

Q81. What are Karst landforms?

Answer: Landforms created by underground water in limestone or soluble rocks through chemical weathering and erosion, such as caves, stalactites, stalagmites, sinkholes and underground rivers.

Q82. State two causes of landslides.

Answer: Natural: heavy and continuous rainfall, earthquakes, steep slopes. Human: deforestation, mining, road construction, unplanned construction and poor drainage.

Q83. What causes avalanches?

Answer: Sudden instability of snow on steep slopes due to heavy snowfall, a sudden rise in temperature, strong winds, earthquakes or human activities like skiing, trekking and construction.

Q84. What causes dust storms?

Answer: Strong winds lifting loose, dry soil and sand, helped by drought, low rainfall, sparse vegetation (deforestation, overgrazing, poor farming) and extreme weather due to climate change.

G. Long Answer Questions (5 marks each)

Q85. Describe the interior of the Earth.

Answer: The Earth has three main layers.

1. **Crust:** the outermost layer; about 30-40 km under continents (continental crust, avg. 30 km) and 5-7 km under oceans (oceanic crust, about 5 km).
2. **Mantle:** a very thick (about 2900 km), hot and mostly solid layer between the crust and the core.
3. **Core:** the innermost, extremely hot and heavy layer. The outer core (about 2200 km) is fluid, mainly iron and nickel. The inner core (about 1250 km) is a solid, hot, spinning metal ball, the densest part of the Earth.

The crust and upper mantle form the rigid lithosphere (about 100 km), which is broken into plates. Beneath it lies the semi-molten asthenosphere (about 200 km), which allows the plates to move.

Q86. Explain the theory of plate tectonics and the three types of plate boundaries.

Answer: Plate tectonics (W.J. Morgan) says the outermost layer of the Earth is broken into tectonic plates that move a few centimetres a year over the semi-molten layer below, driven by convection currents in the mantle.

1. **Convergent boundary:** plates move towards each other. Collision of continental plates forms fold mountains like the Himalaya. When an oceanic plate meets a continental plate, it sinks beneath, causing volcanoes and earthquakes.
2. **Divergent boundary:** plates move apart; magma rises and forms new crust, creating features such as the Mid-Atlantic Ridge.
3. **Transform boundary:** plates slide past each other without creating or destroying crust, mainly causing earthquakes, e.g. San Andreas Fault.

Plate movement forms mountains, valleys, ocean basins, volcanoes and earthquakes, and explains the distribution of continents and oceans.

Q87. 'Plate movements are responsible for the distribution of earthquakes and volcanoes.' Explain.

Answer: Most earthquakes and volcanoes occur along plate boundaries, where plates collide, separate or slide past each other.

- At convergent boundaries, collision or subduction of plates causes strong earthquakes and volcanic activity.
- At divergent boundaries, magma rises and forms new crust.
- At transform boundaries, plates grind past each other causing earthquakes.

The map of earthquakes and volcanoes matches the plate map. The best example is the Ring of Fire around the Pacific Ocean, where the Pacific Plate meets neighbouring plates. Knowing this helps identify earthquake- and volcano-prone regions and manage disasters.

Q88. What is weathering? Explain its types and importance. How is it different from erosion?

Answer: Weathering is the breaking down of rocks on the Earth's surface into smaller pieces, without movement of the material.

1. **Physical:** due to temperature changes, frost or wind.
2. **Chemical:** minerals change through reaction with water, air or acids.
3. **Biological:** plants, animals or micro-organisms, e.g. roots splitting rocks.

Weathering shapes the surface and forms soil.

Difference: erosion involves the removal and transport of soil and rocks by water, wind, ice or waves, while weathering only breaks them down in place.

Q89. Describe the landforms made by running water along the river's course.

Answer: Rivers erode, transport and deposit.
- **Upper course:** steep gradient and strong erosion create V-shaped valleys, waterfalls and rapids. Waterfalls form where hard rock resists erosion and soft rock below wears away; they give tourism, hydroelectric power and recreation.
- **Middle course:** the river meanders, forming oxbow lakes and floodplains as it loses energy and deposits sediment. Fertile soil supports farming.
- **Lower course:** the river slows and deposits large amounts of sediment to form deltas, levees and alluvial fans. Deltas have rich alluvial soil, support rice and jute, fishing and trade, but are flood-prone, e.g. the Sundarbans.

Q90. Explain the landforms created by glaciers and their importance to humans.

Answer: Glaciers scrape and carry huge amounts of material.
Erosional landforms: U-shaped valleys (widened, deepened river valleys), cirques (bowl-shaped depressions at the head of a glacier), arêtes (sharp ridges between valleys), hanging valleys (where smaller glaciers meet larger ones) and fjords (deep, narrow inlets formed when the sea floods glacial valleys).
Depositional landforms: moraines (lateral, medial and terminal) made of till.
Importance: tourism, trekking and skiing; fjords for harbours and fishing; fertile soil for agriculture; moraine dams and lakes for water supply, irrigation and hydroelectric power; and glaciers feed rivers with fresh water.

Q91. Explain the work of wind and waves in shaping the Earth's surface.

Answer: Wind: in dry areas it forms yardangs (streamlined ridges), ventifacts (rocks polished by sandblasting), deflation hollows or blowouts and desert pavements. It deposits sand as dunes: barchan, longitudinal, star and parabolic. These influence settlement and agriculture in arid regions, attract tourists, and dunes act as barriers against desertification.
Waves: they form beaches and sand bars by deposition and cliffs, wave-cut platforms, caves, arches and stacks by erosion. Beaches support tourism and fishing and protect coasts from waves; some coasts need protection to safeguard settlements.

Q92. Describe the four landform-related disasters mentioned in the chapter: landslides, avalanches, GLOFs and dust storms.

Answer: Landslides: caused by heavy rain, earthquakes, volcanic eruptions, steep slopes and loose rocks, plus deforestation, mining, road building and poor drainage.
Avalanches: sudden sliding of unstable snow on steep slopes due to heavy snowfall, sudden warming, strong winds, earthquakes, or skiing, trekking and construction.
GLOFs: sudden release of water from glacial lakes when natural dams of ice or loose moraine fail, due to rapid glacier melting, heavy rain or snow, or earthquakes, avalanches and landslides; causes destructive floods downstream (e.g. the Chamoli flood, Uttarakhand, 2021).
Dust storms: strong winds lift dry loose soil in deserts and semi-arid regions, worsened by drought, sparse vegetation and climate change.

Q93. How have landforms influenced human history and life?

Answer: - Fertile plains and rivers like the Ganga, Nile, Brahmaputra and Indus gave rise to agricultural societies and early cities.
- Mountains acted as barriers and protectors: the Himalayas shielded India, yet passes like the Khyber allowed cultural exchange.
- Deserts like the Thar limited large settlements but encouraged trade routes such as the Silk Route.
- Coasts and harbours supported trade and cultural contact, helping kingdoms in south India flourish.
- Landforms provide water, fertile soil, tourism and hydroelectric power, but also bring risks such as floods, landslides and earthquakes.

Q94. What measures can protect land from erosion? Mention the ancient Indian practices.

Answer: Soil conservation techniques include contouring (trenches along contour lines), bunding (earthen embankments), terracing (steps on slopes), check dams across small streams, and planting vegetation.

The Sindhu-Sarasvati civilisation used contouring, bunding, terracing, dams and canals. Texts like the Vedas, Krishiparashara, Kautilya's Arthashastra and Vrikshayurveda document such practices; the Arthashastra gives guidelines on land assessment by fertility and productivity. The Zabo system in Nagaland uses earthen bunds on hillslopes, and check dams slow water, trap sediment and recharge groundwater.